

①

$$F(s) = \frac{s^4 + 4s^2 + 3}{s(s^2 + 2)} = \frac{(s^2 + 1)(s^2 + 3)}{s(s^2 + 2)}$$

②

$$Z(s) = \frac{k_0}{s} + k_\infty \cdot s + \frac{2ks}{s^2 + \omega^2}$$

$$k_0 = \lim_{s \rightarrow 0} \cdot \cancel{s} \cdot \frac{s^4 + 4s^2 + 3}{\cancel{s}(s^2 + 2)} = \frac{0 + 0 + 3}{0 + 2} = \frac{3}{2}$$

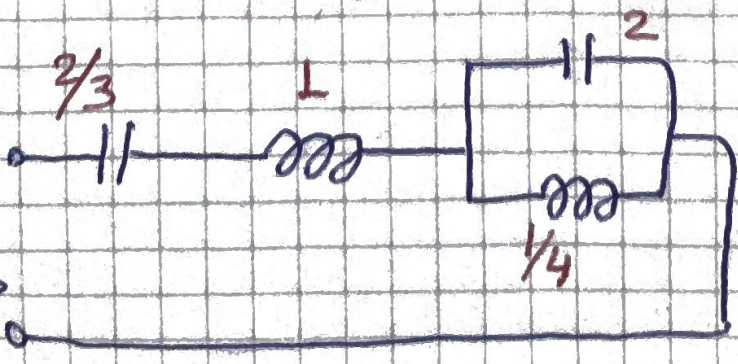
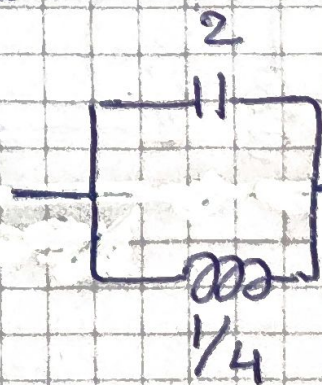
$$k_\infty = \lim_{s \rightarrow \infty} \cdot \frac{1}{s} \cdot \frac{s^4 + 4s^2 + 3}{s(s^2 + 2)} = \lim_{s \rightarrow \infty} \frac{s^4 + 4s^2 + 3}{s^4 + 2s^2} = 1$$

$$2k = \lim_{s^2 \rightarrow -2} \frac{\cancel{(s^2 + 2)}}{s} \cdot \frac{s^4 + 4s^2 + 3}{s \cancel{(s^2 + 2)}} = \frac{(-2)^2 + 4(-2) + 3}{(-2)} = \frac{1}{2}$$

$$Z(s) = \frac{3/2}{s} + s + \frac{1/2 \cdot s}{s^2 + 2}$$

$$\frac{3}{2} \cdot \frac{1}{s} \rightarrow \frac{1}{2/3} \quad \triangleright \quad s \rightarrow \frac{1}{1}$$

$$\frac{1/2 \cdot s}{s^2 + 2} = \frac{1}{\frac{2s^2}{s} + \frac{4}{s}} = \frac{1}{2s + \frac{4}{s}} \rightarrow \frac{1}{2s + \frac{4}{s}}$$



Z(s)

$$\textcircled{b} Y(s) = \frac{k_0}{s} + s k_\infty + \frac{2k s}{(s^2+2)}$$

$$k_0 = \lim_{s \rightarrow 0} s \cdot \frac{s^4 + 4s^2 + 3}{s(s^2+2)} = 3/2$$

$$k_\infty = \lim_{s \rightarrow \infty} \frac{1}{s} \cdot \frac{s^4 + 4s^2 + 3}{s(s^2+2)} = 1$$

$$2k = \lim_{s^2 \rightarrow -2} \frac{(s^2+2)}{s} \cdot \frac{s^4 + 4s^2 + 3}{s(s^2+2)} = 1/2$$

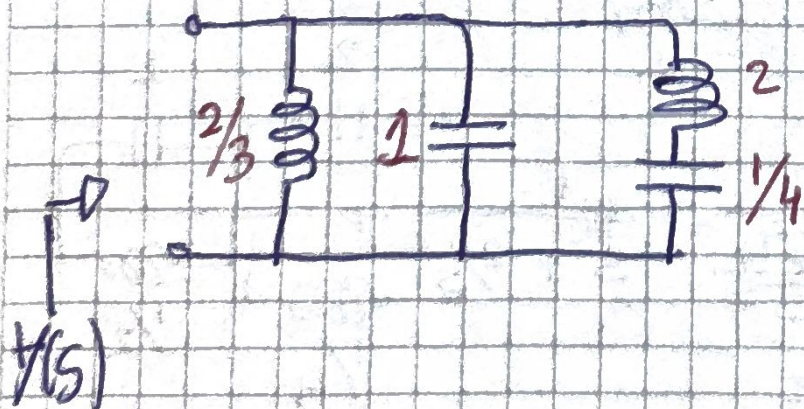
$$Y(s) = \frac{3/2}{s} + s + \frac{1/2 s}{s^2+2}$$

$$\triangleright \frac{3/2}{s} \rightarrow \begin{array}{|c|} \hline \text{---} \\ \hline \text{---} \\ \hline \end{array} \begin{array}{|c|} \hline 2/3 \\ \hline \end{array}$$

$$\triangleright s \rightarrow \begin{array}{|c|} \hline \text{---} \\ \hline \text{---} \\ \hline \end{array} 1$$

$$\triangleright \frac{1/2}{s^2+2} = \frac{1}{2s+4/s} \rightarrow$$

$$\begin{array}{|c|} \hline \text{---} \\ \hline \text{---} \\ \hline \end{array} \begin{array}{|c|} \hline 2 \\ \hline \end{array} \begin{array}{|c|} \hline 1/4 \\ \hline \end{array}$$



(c) $Z(s) = \frac{s^4 + 4s^2 + 3}{s^3 + 2s}$

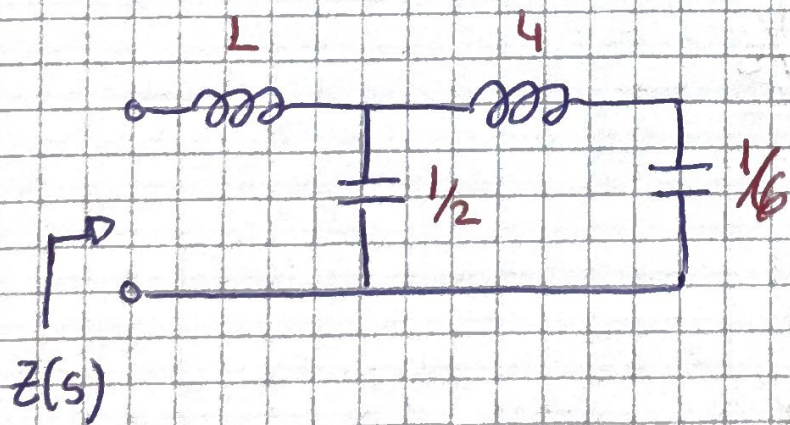
	0	∞
Z	$\frac{1}{SC}$	SL
Y	$\frac{1}{SL}$	SC

$$Z \left\{ \begin{array}{l} S^4 + 4S^2 + 3 \\ - S^4 + 2S^2 \\ \hline 0 + 2S^2 + 3 \end{array} \right. \left| \begin{array}{l} S^3 + 2S \\ S \end{array} \right. \xrightarrow{\quad} \text{Diagram of a mass-spring-damper system}$$

$$\begin{array}{r}
 s^3 + 2s \quad | \quad 2s^2 + 3 \\
 \underline{s^3 + \frac{3}{2}s} \quad \frac{1}{2}s \\
 0 + \frac{1}{2}s
 \end{array}
 \longrightarrow
 \begin{array}{c}
 \text{---} \\
 | \\
 \text{---} \\
 | \\
 \text{---}
 \end{array}
 s^{1/2}$$

$$7 \left\{ \begin{array}{r} 2s^2 + 3 \\ - 2s^2 \\ \hline 0 + 3 \end{array} \right. \begin{array}{r} \frac{1}{2}s \\ 4s \end{array} \longrightarrow \text{---} \bullet \text{---} \text{---} \text{---} \bullet \text{---}$$

$$\left\{ \begin{array}{c} \frac{1}{2}s \\ 0 \end{array} \right\} \left| \begin{array}{c} 3 \\ \frac{1}{6}s \end{array} \right. \longrightarrow \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] \frac{1}{6}$$



① $z(s) = \frac{3 + 4s^2 + s^4}{2s + s^3}$

	0	∞
Z	$\frac{1}{SC}$	SL
X	$\frac{1}{SL}$	SC

$$\begin{array}{r} 3 + 4s^2 + s^4 \quad | \quad 2s + s^3 \\ - \quad 3 + \frac{3}{2}s^2 \\ \hline 0 + \frac{5}{2}s^2 + s^4 \end{array}$$

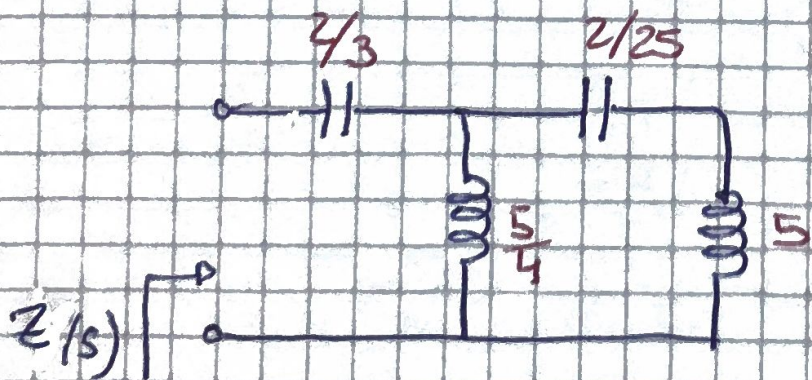
$$\frac{3}{2} \cdot \frac{1}{5}$$



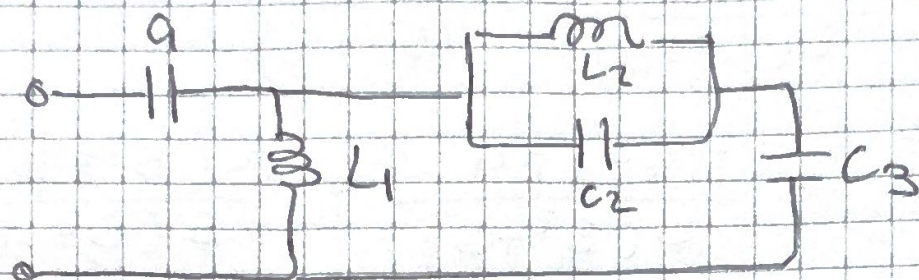
$$\begin{array}{r} 2s + s^3 \mid \frac{5}{2}s^2 + s^4 \\ - \underline{2s + \frac{4}{5}s^3} \quad \frac{4}{5} \cdot \frac{1}{s} \\ \hline 0 + \frac{1}{5}s^3 \end{array} \quad \rightarrow \quad \begin{array}{c} \text{---} \\ | \\ \text{---} \end{array} \quad \frac{5}{4}$$

$$\begin{array}{r} \frac{5}{2}s^2 + s^4 \overline{) \frac{1}{5}s^3} \\ - \frac{5}{2}s^2 \\ \hline 0 + s^4 \end{array} \quad \frac{25}{2} \cdot \frac{1}{5} \rightarrow 0 \quad \frac{1}{1} \quad \frac{2}{25}$$

$\frac{1}{s^3} \mid \frac{1}{s^4}$
 $\frac{1}{s} \cdot \frac{1}{s}$



②
$$Y(s) = \frac{s^5 + 18s^3 + 48s}{6s^4 + 42s^2 + 48}$$



$$Z(s) = \frac{6s^4 + 42s^2 + 48}{s^5 + 18s^3 + 48s} = \frac{48 + 42s^2 + 6s^4}{48s + 18s^3 + s^5}$$

Z {
$$\begin{array}{r|l} 48 + 42s^2 + 6s^4 & 48s + 18s^3 + s^5 \\ \hline 48 + 18s^2 + s^4 & \frac{1}{s} \end{array}$$
 $\rightarrow$ C_1

$Z_2 = \frac{24s^2 + 5s^4}{48s + 18s^3 + s^5} \rightarrow Y$

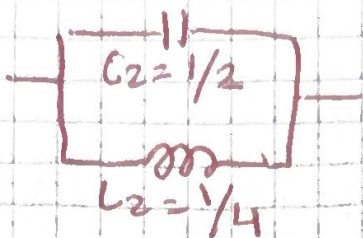
Y {
$$\begin{array}{r|l} 48s + 18s^3 + s^5 & 24s^2 + 5s^4 \\ \hline 48s + 10s^3 & 2 \cdot \frac{1}{s} \end{array}$$
 $\rightarrow$ $L_1 = 1/2$

$$Y_4 = \frac{8s^3 + s^5}{24s^2 + 5s^4} = \frac{s^5 + 8s^3}{5s^4 + 24s^2} = \frac{\cancel{s^2}(s^3 + 8s)}{\cancel{s^2}(5s^2 + 24)}$$

$Z_4 = \frac{5s^2 + 24}{s(s^2 + 8)} \rightarrow$ $= \frac{2ks}{(s^2 + \omega^2)}$

$$2k = \lim_{s^2 \rightarrow -8} \frac{\cancel{(s^2 + 8)}}{s} \cdot \frac{5s^2 + 24}{s\cancel{(s^2 + 8)}} = \frac{5(-8) + 24}{(-8)} = 2$$

$$\frac{25}{(s^2+8)} = \frac{1}{\frac{1}{2}s + 4 \cdot \frac{1}{s}}$$



$$Z_0 = Z_4 - \frac{25}{(s^2+8)} = \frac{5s^2+24}{s(s^2+8)} - \frac{25}{(s^2+8)}$$

$$Z_0 = \frac{5s^2+24-25s^2}{s(s^2+8)} = \frac{3s^2+24}{s(s^2+8)} = \frac{3(s^2+8)}{s(s^2+8)}$$

$$Z_0 = \frac{3}{s} \rightarrow \text{Circuit diagram of a capacitor } C_3 = 1/3$$

